

Capstone Project Documentation

Project Title: Optimizing Inventory Management and Sales Forecasting Through Data Insights

Objective: To analyze supermarket sales, inventory, customer behavior, and supplier data to derive actionable insights that improve inventory management, sales forecasting, pricing strategies, and supply chain efficiency.

Tools Used:

- Microsoft Excel (Data Cleaning, Pivot Analysis)
- MySQL Workbench (Data Storage, Querying & Transformation)
- Power BI (Data Visualization & Dashboard Creation)

Dataset Description: A simulated Walmart Sales Forecasting dataset was used, enriched to include the following fields:

- Store
- Date
- Weekly_Sales
- Temperature
- Fuel_Price
- CPI (Consumer Price Index)
- Unemployment
- Inventory_Level (Simulated based on units sold + buffer)
- Supplier_ID (Simulated)
- Customer_Type (Regular, New, Premium - Simulated)
- Product_Category (Simulated)
- Units_Sold
- Unit_Price

Step 1: Data Cleaning & Preprocessing (Excel)

- Checked for missing values and corrected inconsistencies.

- Added simulated fields (Inventory Level, Supplier ID, Customer Type, Product Category, Unit Price).
- Removed unnecessary columns.
- Saved the cleaned data as an enriched Excel file.

Pivot Analysis in Excel:

- Sales by Product Category
 - Inventory by Store
 - Units Sold by Customer Type
 - Average Unit Price per Product Category
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Step 2: SQL Queries in MySQL Workbench

Table Created:

```
CREATE TABLE walmart_data (  
    Store INT,  
    Date DATE,  
    Weekly_Sales DECIMAL(10,2),  
    Temperature DECIMAL(5,2),  
    Fuel_Price DECIMAL(5,2),  
    CPI DECIMAL(10,2),  
    Unemployment DECIMAL(5,2),  
    Inventory_Level INT,  
    Supplier_ID VARCHAR(10),  
    Customer_Type VARCHAR(10),  
    Product_Category VARCHAR(50),  
    Units_Sold INT,  
    Unit_Price DECIMAL(6,2)  
);
```

SQL Queries Written:

1. Total Sales by Store
2. Monthly Sales Trend
3. Product Category-wise Sales
4. Average Inventory by Store
5. Customer Type Purchase Behavior
6. Top Suppliers by Sales
7. Unit Price vs Units Sold (Top Products)
8. Quarterly Sales Overview

These queries extracted key insights about sales distribution, seasonal patterns, supplier performance, and customer behavior.

Step 3: Power BI Dashboard Creation

Dashboard Pages:

1. **Overview** – KPIs like Total Sales, Avg. Unit Price, Total Units Sold
2. **Store Analysis** – Sales and Inventory by Store
3. **Customer Insights** – Sales by Customer Type, Tree Map (Customer Type vs Product Category)
4. **Product Insights** – Category-wise Sales, Avg Unit Price
5. **Supplier View** – Top 5 Suppliers by Sales
6. **Seasonality View** – Monthly/Quarterly Sales Trend

Extra Visuals Added:

- Stacked Column Chart: Customer Type over Time
- Tree Map: Customer Type vs Product Category

Color Theme:

- Primary Colors: Walmart Blue (#0071ce), Yellow (#ffc220), Light Gray, White

Key Findings:

- Premium customers contribute to higher revenue despite lower volume.
- Specific product categories drive most sales (e.g., Groceries, Apparel).
- Inventory levels vary widely; some stores are overstocked.
- Certain suppliers consistently outperform others.
- Seasonality influences purchasing behavior significantly.